

Contact

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Google Scholar: profile
Doctoral dissertation: available here

Expertise & Skills

Data engineering and analysis
- Azure, ADF, Databricks, Microsoft Fabric, PySpark, Python, SQL, Snowflake, MATLAB, Crosser, ADE, Docker, git, Azure DevOps

Modeling & Simulation
- FEM (COMSOL)
- MATLAB & Simulink

Other
- MCP Architecture, Azure AI Search
- Miro, Obsidian+Excalidraw
- Bash, PowerShell, MS Office, LaTeX, Markdown
- control engineering, microfluidics

Language

Finnish: Mother tongue
English: C2 (full professional working proficiency)
Swedish: A2
German: A1
Mandarin: A1

Certifications

More information in Credly

Databricks
Databricks Certified Data Engineer Associate

Agile Data Engine
Agile Data Engine for Data Engineers

AWS
AWS Certified Cloud Practitioner

Azure
Azure AI Fundamentals (AI-900)
Azure Data Fundamentals (DP-900)
Azure Fundamentals (AZ-900)

Scrum
Professional Scrum Master I (PSM I)

ANTTI-JUHANA MÄKI

SENIOR DATA ENGINEER, D.SC. (TECH.)

About Me

Passionate and experienced data guy with a strong mathematical modeling and control engineering background. Building resilient data solutions as a Senior Data Engineer at Siili. Main tools are **Azure** (especially **ADF** and **Azure DevOps**), **Databricks** (PySpark), **Microsoft Fabric**, **Python**, and **SQL**.

Experience

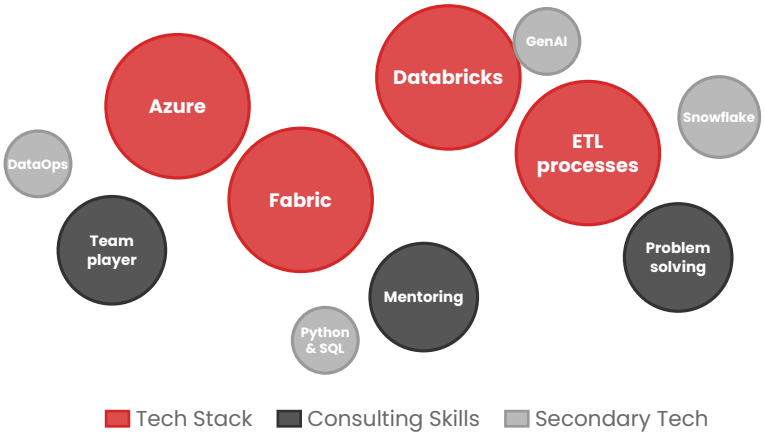
Senior Data Engineer, Siili 2026/02 – Present
Working primarily in Azure, Databricks, Microsoft Fabric, and Snowflake. A growing focus on GenAI topics including MCP architecture, ontology-driven AI systems, and semantic search, emphasizing data quality and governance.

In-house Accelerator Tech Lead & Architect, Solita 2025/09 – 2026/01
Led an in-house agent-based (MCP server) innovation and development team of seven members, serving as Technical Lead, Project Manager, and Scrum Master while collaborating closely with cross-functional stakeholders. Designed the MCP Reference Architecture for internal GenAI accelerators.

Data Engineer, Solita 2023/03 – 2026/01
Worked mainly in Azure. Customers were private companies (energy and Industrial IoT sectors), Football Association of Finland, and Public sector organizations. Also worked with Snowflake, Crosser, git, Docker, ADE, and Microsoft Fabric.

Data Engineer, Futurice 2022/10 – 2023/02
Intensive five months of learning (e.g. 5 certs), and customer work in retail industry, handling their operational data using Azure (especially ADF), PostgreSQL, Databricks, and Python.

Academia 2006 – 2022
Postdoctoral Research Fellow, Tampere University 2018/09 – 2022/10
Data analysis methods for challenging biosignals; Vice-PI in a multidisciplinary team and lead of a researcher group developing a vascular-on-a-chip platform. Before that Doctoral Researcher at TUT (2011–2018, D.Sc. 2018): control systems and mathematical modeling. More info on my website: anahill.github.io/academic.html



Selected Projects

Data Engineering Services 2026/04 – 2026/08
Private Company, Finland (NDA) · Role: Lead Senior Data Engineer · Industry: Food Industry

- Designing and building a scalable **data platform** in Microsoft Fabric
- Lead developer with architectural responsibilities, leading a small development team
- Consolidated a fragmented set of orchestration pipelines into one reliable production pipeline
- Authored the team's technical documentation: architecture guides, naming conventions, and an interactive column-mapping reference

Tech stack: Microsoft Fabric (Lakehouse, Warehouse, Data Pipelines, Notebooks, DAGs, PySpark/Delta Lake), Azure DevOps

GenAI Operational Accelerator 2025/09 – 2026/01
Solita (In-house Project), Finland · Role: Tech Lead / GenAI Solutions Architect / Project manager / Scrum master

- Led in-house **agent-based (MCP server) innovation and development efforts**, serving as Technical Lead, Project Manager, and Scrum Master
  - Team comprised of experienced software developers, a cloud engineer, and an IT specialist
- Designed and established the **MCP Reference Architecture** for internal GenAI accelerators
- Collaborated closely with multiple in-house stakeholders across business and technology domains

Tech stack: Azure, Azure AI Search, Azure DevOps, Miro, Obsidian & Excalidraw

Data Engineering Services 2024/03 – 2025/06
Private Company, Finland (NDA) · Role: Data Engineer · Industry: Energy Sector

- Building and maintaining a scalable **data platform** spanning both **data lakes** and **relational databases**.
- Optimizing **data pipelines** to enhance performance and reliability.
- Ensuring seamless **data integration** across multiple sources for analytics and operations.
- Working with **time-series** data

Tech stack: Databricks, PostgreSQL, Azure, Azure Data Factory, Azure DevOps

Developing Machine Learning Capabilities 2023/05 – 2024/04
Public sector organization in Finland (NDA) · Role: Machine Learning Engineer and Data Scientist · Industry: Public Sector

- Developing machine learning capabilities for the customer, using both supervised and unsupervised learning, created for example a supervised classification algorithm using Databricks for the customer ERP system.

Tech stack: Databricks, AutoML, Snowflake, SQL, Python & Spark

All projects on my website: anahill.github.io/work.html#projects

Education

Doctor of Science (Technology) 2018/09
Tampere University of Technology, Tampere, Finland
title: "Modeling and control of microscale cell culture environments", available here.

Master of Science (Technology) "with distinction" 2010/07
Tampere University of Technology, Tampere, Finland
Major: Automation and Control

Student exchange 2006-2007
Hong Kong Polytechnic University

Matriculation Examination 2002
Seinäjoki Upper Secondary School

Merits & Hobbies

Solita Connected Data Clippy Award 2024
- For a person who is always helping and sharing knowledge

Invention disclosures
- TKS0222021 MEA-signaalin analyysityökalu (“MEA analysis tool”), granted 05/2021
- TKS0612020 Pilarivapaa 3D monisolukasvatuschippi (“Barrier-free 3D co-culture chip”), granted 06/2020

Member of the Organizing Committee
Nordic Organ on a Chip Symposium 08/2019

Finnish military rank
Second lieutenant

Hobbies
Football, reading, cross country skiing, orienteering